

Exercise B

In the linear model $E[y_i] = \beta_1 + \beta_2 x_i$, suppose that instead of observing x_i we observe $x_i^* = x_i + u_i$, where u_i is independent of x_i and $\text{var}(u_i) = \sigma_u^2$, with $E[u_i] = \mu_u$. Recall that

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i \quad (\text{By assumption})$$

Add and subtract $\beta_2 u_i$

$$\begin{aligned} &= \beta_1 + \beta_2 u_i - \beta_2 u_i + \beta_2 x_i + \varepsilon_i \\ &= \beta_1 + \beta_2 (x_i + u_i) - \beta_2 u_i + \varepsilon_i \end{aligned}$$

Add and subtract $\beta_2 \mu_u$

$$= \beta_1 + \beta_2 \mu_u + \beta_2 x_i^* + \underbrace{\beta_2 (\mu_u - u_i)}_{\varepsilon_i^*} + \varepsilon_i$$

$$y_i = \beta_1^* + \beta_2^* x_i^* + \varepsilon_i^*$$

$\varepsilon_i^* \stackrel{\text{iid}}{\sim} \mathcal{N}_*$, with

$$\beta_1^* = \beta_1 + \beta_2 \mu_u; \quad \beta_2^* = \beta_2.$$

$$\begin{aligned} E[\varepsilon_i^*] &= \beta_2 E[\mu_u - u_i] + E[\varepsilon_i] = 0 \\ \text{var}(\varepsilon_i^*) &= \beta_2^2 \sigma_u^2 + \sigma^2 = \sigma_*^2 \end{aligned}$$

This is a linear model with covariate x_i^* and errors ε_i^* . The estimator

$$\hat{\beta}_2^* = \frac{\text{cov}(x^*, y)}{\text{var}(x^*)} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i^* - \bar{x}^*)(y_i - \bar{y})}{\frac{1}{n} \sum_{i=1}^n (x_i^* - \bar{x}^*)^2}$$

is unbiased and consistent for $\beta_2^* = \beta_2$, even under a "perturbed" covariate.

$$E[\hat{\beta}_2^*] = \beta_2^* = \beta_2.$$

(This follows from the equations above, but you can prove it directly).

However, the estimator for β_1 is problematic

$$E[\hat{\beta}_1^*] = E[\bar{y} - \bar{x}^* \hat{\beta}_2^*] = \boxed{\beta_1^* \neq \beta_1}$$

because $\beta_1^* = \beta_1 + \beta_2 \mu_u$.

Therefore the estimator is inconsistent for β_1 unless

1. $\beta_2 = 0$ ($\beta_1^* = \beta_1 + 0 = \beta_1$). The covariate is irrelevant
a non-interesting situation.

2. $\mu_0 = 0$ ($\beta_1^* = \beta_1 + 0 = \beta_1$). The measurement error is
unbiased.